Title

Your Name

3. Mai 2026

Abstract

Abstract text.

Zusammenfassung

Abstract text.

1 Nomenclature

1. SKA: Square Kilometer Array
2. EoR: Epoch of Reionization
3. SBI: Simulation-Based Inference
4. CNN: Convolutional Neural Network
5. MLP: Multilayer Perceptron
6. KL: Kullback-Leibler divergence
7. SM: Standard Model
8. WDM: Warm Dark Matter
9. ΛCDM: Lambda Cold Dark Matter
10. IGM: Intergalactic Medium
11. CLT: Central Limit Theorem
12. MI: Mutual Information
13. CE: Conditional Entropy

2 Introduction

Give the cosmological setting, small intro to the EoR, and the 21 cm signal. Then introduce the problem of parameter inference, and how it is usually done. Then introduce the idea of using machine learning for this problem, and how it can be done in a more interpretable way.

The SKA and why simulations are needed.

SBI in general and why it is needed in this field.

Latent space and geometry.

3 Physics Background

A few topics: - short introduction to history of universe - then focus on EoR - 21 cm signal from QM and how it can be used to learn about the EoR

Still need some more subsections to explain the other parameters being examined

3.1 The Cosmological Setting

For the purposes of this thesis, the standard cosmological model and an additional warm dark matter parameter are assumed.

Also assuming inflation..., though this not discussed here.

The standard cos

Cite a few papers that give the current status of cosmology in ΛCDM, e.g. Planck 2018, etc.

The ΛCDM-model assumes:

The six parameters describing the ΛCDM-model and their estimated values today are shown in table 1.

These are not the fundamental params. Choose which exactly to include.

Need today's parameters so that can explain choice of priors

Do I want a derivation of these here?

Table 1: Parameters of ΛCDM with estimated values

Name	Symbole	Value	Unit	Source
Hubble constant	H_0	67.4	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$	Planck 2018
Matter density	Ω_m	0.315 ± 0.007		Planck 2018
Dark energy density	Ω_Λ	0.685		Planck 2018
Spectral index	n_s	0.965		Planck 2018
Matter fluctuation amplitude	σ_8	0.811 ± 0.006		Planck 2018
Equation of state	w	-1		Planck 2018

Under these assumptions, one can solve Einstein's field equations. The solutionss are the Friedmann equations:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} - \frac{k}{a^2}, \tag{1}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3} \tag{2}$$

The Friedmann equations describe the evolution of the universe starting at approx THIS time.

Could define redshift here

According to the cosmological standard model, the timeline of the universe is approximately as follows:

The first phase of the universe is marked by *radiation domination*. Immediately after the Big Bang, the universe underwent a period of rapid expansion called *inflation*, during which space expanded by appriximately an order of 10^{26}. This inflation was driven by the so-called inflaton field. During the following process of reheating, the energy of the inflaton field was converted to SM particles, which enabled subsequent *baryogenesis*, i.e. the creation of matter and antimatter particles. In the subsequent annihilation of matter.antimatter pairs, a small excess of matter paticles survived, composing the matter content of the universe today. There have been many proposed mechaninisms for both reheating and baryogensesis, and both are active fields of research. Insert sources here In the following time scale of $20 \cdot 10^{-12}\mathrm{s} - 1000\mathrm{s}$, particles acquired mass and coalesced into hadrons. Neurinos decoupled from baryons and thus the cosmic neutrino background was formed. What followed was an epoch named *big bang nuceosynthesis*, during which mostly protons and neutrons bound to primordial nuclei, consisting mostly of hydrogen, Helium-4 and traces of deuteron, Helium-3 and lithium.

The second phase of the universe is marked by *matter domination*. During *recombination*, electrons and nuclei where bound to neutral atoms. Photons were no longer in thermal equilibrium with matter and thus ”decouple”, forming the cosmic microwave background (CMB). Measurements of the CMB today provide one of the most crucial constraints on cosmological parameters.

After recombinations, most photons propagate freely and are quickly redshifted, such that the universe becomes dark. This period is therefore called the *dark ages*. Matter has not yet coalesced into stars and galaxies. The main source of photons visible phtons? is the 21 cm emission line of neutral hydrogen.

Once first stars and galaxies have formed, they emit photons energetic enough to ionize the surrounding hydrogen. This period is called the *epoch of reionization* (EoR). By the end of the EoR, most of the IGM is ionized and continues to be so today.

These sections are based on [1]

Maybe short example of how an evolution might be described...

How other cosmological models can be tested with the EorFlow setup, maybe also put this somewhere else.

What is matter distribution at the redshift when beginning study?

Brief summary of CMB

Small overview of existing studies of timeline of universe, why there is a gap in EoR

Maybe cite more recent paper that explains why studying this exact regime of redshifts

3.2 The Astrophysical Setting

3.3 On the Origin of the 21 cm Signal

Temperature of the ISM at around these redshifts, why neutral hydrogen does not absorb of emit visible light at these temperatures.

How trace elements interact with visible light at these temperatures.

At least mention the possibility of a derivation

Quantum mechanical aspect: hyperfine splitting of ground state of hydrogen.

Statistical mehanics aspect: spin temperature and how it is useful in measuring the 21 cm signal.

- Einstein coefficients from QM - Different processes that influence the spin temperature, e.g. collisional coupling, Wouthuysen-Field effect, etc. - Final formula for spin temperature

3.4 Evolution of the 21 cm Signal

4 Simulations

Reason for SBI

Could either put standard cosmology here or into previous section. Need to explain params used in simulations.

21 cm fast

The parameters used, and the priors on them. Why these parameters? What do they mean physically?

The six parameters and their priors chosen for the simulations in this project are summarized in table 2. The priors are chosen to be uniform and constrained to a range motivated by multimessenger observations.

Table 2: Parameters used for the simulations

Name	Symbol	Prior
Matter density	Ω_m	$\mathcal{U}(0.2, 0.4)$
Warm dark matter mass	ω_{WDM}	$\mathcal{U}(0.3, 10)$ keV
Minimum virial temperature	T_{vir}	$\mathcal{U}(10^4, 10^{5.3})$ K
Ionization efficiency	ζ	$\mathcal{U}(10, 250)$
Specific X-ray luminosity	σ_8	$\mathcal{U}(10^{38}, 10^{42})$ erg/s/M$_\odot$ $\times$ yr
X-ray energy threshold for self-absorption by host galaxies	E_0	$\mathcal{U}(100, 1500)$ eV

interpretation of data

Noise simulations

If in end used different types of data, then explain them here. And why less params where used for information geometry.

How reioniazation history is calculated

5 Information Theory and Neural Estimators

Information theory: - Entropy, KL divergence, mutual information - Interpretation of these quantities, maybe expectation?

Information geometry: - Fisher information metric, geodesic distance, curvature - Interpretation of these quantities, maybe expectation?

5.1 Some Basic Measures in Information Theory

Given a discrete random variable $X \sim p$, the entropy of X is defined as:

$$\mathbb{H}(X) = -\sum_x p(x) \log p(x) = -\mathbb{E}_X \left[\log p(X)\right] \geq 0. \tag{3}$$

If the logarithm is chosen as with base 2, one defines the entropy to be measured in units of bits. If one uses the natural logarithm, the entropy is measured in units of nats.

Entropy is a measure of the unvertainty of a random variable. A high entropy corresponds to a lack of predictability and as such to a high amount of information contained in the distribution.

As an example, the entropy of the uniform distribution is maximal, while that of the delta distribution is zero.

Do better with the example.

Another important metric in information theory is the Kullback-Leibler divergence, which can be interpreted as a measure of distance between to probability distributions.

X and Y might be same? At least p and q need to be defined on same probability space.

Let $X \sim p$ and $Y \sim q$ be two discrete random variables. The KL divergence between

p and q is defined as:

$$\mathbb{D}_{\mathrm{KL}}(p||q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_X \left[\log p(X) - \log q(X)\right] = \mathbb{H}(X) + \mathbb{E}_X \left[-\log q(X)\right] \tag{4}$$

The KL-divergence is not symmetric and does not satisfy the triangle inequality and as such is not a metric. However, it does still continue

Lastly, the mutual information between two random variables X and Y is defined as: Change some of these to Y, this is still autocorrect

$$\begin{aligned}
\mathbb{I}(X;Y) &= \mathbb{D}_{\mathrm{KL}}(p(x,y)||p(x)p(y)) && (5)\\
&= \mathbb{H}(X) - \mathbb{H}(X|Y) && (6)\\
&= \mathbb{H}(Y) - \mathbb{H}(Y|X). && (7)
\end{aligned}$$

How mutual information quantifies the amount of information obtained about a distribution when knowing another.

For discrete random variables, one can write:

$$\mathbb{I}(X;Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \tag{8}$$

which makes a Monte-Carlo estimation of mutual information accessible.

Maybe something on limitations on the interpretation of these metrics. Could also tease limitations in high dimensions, or give intuition for amount of data needed.

Something on summary statistics?

6 Basic Neural Networks

EoRFlow general description, goal etc. Why it is needed. Have figure of architecture.

How this type of model can be used in 21 cm cosmology in general, and then how it is used here specifically.

Traditionally, data as complex as those expected from SKA are analyzed using summary statistics. A summary statistic is a function that maps the data to a lower-dimensional space, typically with some physical interpretation. An example is the power spectrum. The interpretation of the power spectrum is blablabla. However the power spectrum is not a sufficient statistic for the 21 cm signal. As an alternative, one can use a neural network to learn a summary statistic of the data. While being potentially more informative, this comes at the cost of less interpretability.

For cnn and flow: - general stuff on model - architecture, where it's inspired from, how it was adapted to the problem, etc - loss function - backpropagation

Training - details on scheduler, optimizer, etc - training hyperparameters, chip used

Why is a generative model needed? Why is the flow chosen? What are the advantages of flows over other generative models?

Why this very specific architecture?

Do I want a short ML intro? Could e.g. introduce MLP here and rename section to NNs in general

A word about how architectures will be discussed, e.g. notation with h, z, x, etc.

6.1 Convolutional Neural Networks

CNNs are inspired by the task of implementing neural networks in image analysis, i.e. on data with 2D spatial structure. An early version was successfully implemented in 1983 with Fukushima's "Neucognitron" Neocognitron citation. An architecture implementing backpropagation was LeCun's "LeNet" LeNet citation.

To explain the novel concepts behind the CNN in comparison with a MLP, the CNN architecture will first be discussed in 1D. The generalization to higher dimensions will be discussed afterwards.

Let X denote the data space and Y the feature space.

Let the input have dimensions $x \in \mathbb{R}^{C \times H \times W}$ where C the channel dimension, H the height dimension and W the width dimension. In case of an RGB image, this channel dimension would be three.

The CNN learns a function $f_\theta : X \to Y$.

As opposed to a simple MLP, the CNN offers the following advantages:

Using summation convention

Not writing bias explicitly

A basic MLP consists of only one type of layer. If h is the output of the previous layer, the current layer is defined by

$$z_i = \phi(W_{ij}h_j), \tag{9}$$

The CNN implements additional layers.

A convolutional layer applies a convolution operaton:

Talk abt convolution vs cross-correlation here

Is this def even correct? Not need different indices on h?

$$z_i = \phi([w \circledast h]_i) = \phi(\sum_k w_k h_{i+k}) \tag{10}$$

where the dimensionality of the kernel $w \in \mathbb{R}^K$ is typically much lower than that of the data.

A pooling layer applies a pooling operation:

$$z_i = \operatorname{pool} K_{ij}h_j \tag{11}$$

where K defines a pooling window for each index i. Typical choices for pool are maxpool, which chooses the maximum of all given values, or meanpool, which calculates their mean.

The purpose of a pooling layer is to...

Add normalization layers here

Add info on stride params etc here

Put it together here, potentially add image of total architecture.

6.2 Normalizing Flows

Choose a Schreibweise. Use p_u if necessary, otherwise $u \sim p(u)$?

Differentiate the true target distribution and the modelled one. Then going for Kullback Leibler of the two is equivalent to maximizing log likelihood.

A normalizing flow model is a generative model that learns to transform a simple (often Gaussian) distribution into a more complex one. The transformation is a composition of invertible functions, which are chosen to be computationally efficient to invert.

Let $u \sim p_u = \mathcal{N}(0, I_n)$ with $n \in \mathbb{N}$ the base distribution and $x \sim p_x$ how to signal that x is n-dimensional? the target distribution.

The normalizing flow learns a transformation given by an invertible function $f : \mathbb{R}^n \to \mathbb{R}^n$, such that sampling from the base distribion and applying the transfomation yields samples from the target distribution.

Let $f : \mathbb{R}^n \to \mathbb{R}^n$ a differentiable, invertible function with $g = f^{-1}$ its inverse (i.e. f is a diffeomorphism). Then the model is a normalizing flow if:

$$x = f(u) \sim p_x, \quad u \sim p_u. \tag{12}$$

Should have two sim signs, no? How to differentiate sampling and being part of distribution?

Once the flow is learned, the process to generate samples from p_x is $u \sim p_u$ and subsequent $x = f(u)$.

In order to compute $p_x(x)$ for a given x, on applies the change of variables formula:

$$p_x(x) = p_u(g(x)) \left| \det \left(\frac{\partial g(x)}{\partial x} \right) \right| = p_u(u) \left| \det \left(\frac{\partial f(u)}{\partial u} \right) \right|^{-1}. \tag{13}$$

The normalizing flow becomes expressive by allowing f to be as complex as possible, while restricting f to a certain class of functions with efficiently computable Jacobian determinants makes it computationally tractable.

Come up with nicer notation for $g_i(\cdot|\theta_i)$

In practice, one often models g as a composition of simple neural networks, e.g. multilayer perceptrons. Let $g_\theta = g_1 \circ g_2 \circ ... \circ g_k$ with $g_i : \mathbb{R}^n \to \mathbb{R}^n, x \mapsto g_i(x|\theta_i)$ and $k \in \mathbb{N}$.

Now the loss becomes

$$\text{NLL}_\theta = \sum_{i=1}^{k} \log|\det J(g_i)| - \log p_u(g_\theta(x)) \tag{14}$$

Now what this specific Freia flow does. Research on website!

Or maybe put exact architecture in appendix?

6.3 Variational Autoencoders

7 Specific Architectures

The following setup is common in current 21cm SBI:

One networks learns a compression of the image data to a much lower-dimensional latent space, i.e. a summary statistic. Another network learns a mapping from the latent space to the parameter space.

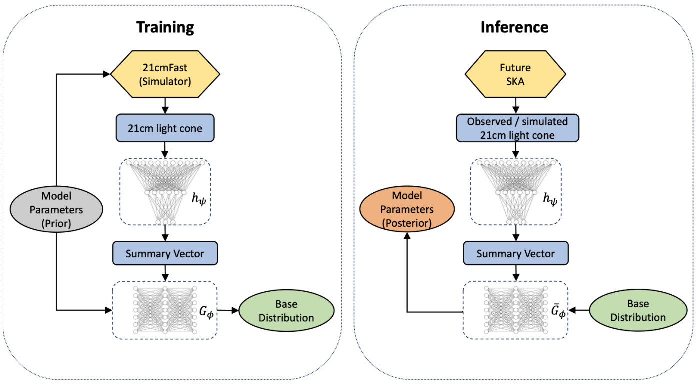
The compressing network usually has a much heavier architecture. Once it has been trained, one can learn different lighter networks to map its summary to different parameter spaces. Example for parameter spaces include the parameters used for the simulations, or a simple physical summary statistic of the image data.

Maybe put some more examples, e.g. SKATR?

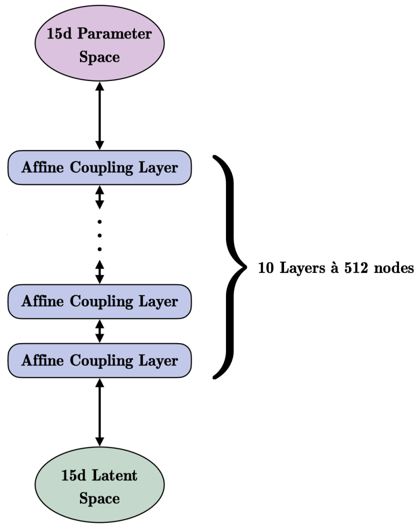
Which construction is PIE-INN based on? Is it completely new?

One such construction is *21cmPIE-INN*, which consists of a CNN acting as a compressing network and a normalizing flow mapping its output to six parameters used for the original simulations. The architecture is visualized in figure 1a.

It is also possible to learn e.g. the reionization history with variable resolution from the CNN's summary, as is done with the *EoRFlow* architecture Citation here. The flow implemented in EoRFlow is slightly more complex the the 21cmPIE-INN. It's architecture is visualized in figure 1b.



(a) Architecture of 21cmPIE-INN, taken from citation



(b) Architecture of the flow used in EoRFlow, taken from citation

Figure 1: Overall caption for both figures

8 Implementation

Mutual information: - General, then specific formula used here - Interpretation, maybe expectation?

How interpretable is MI? What can really be understood from it?

Why is it useful in a physics setting? What can it tell us about the physics of the problem?

Should I be using past tense here?

For training, an ensemble of models was training for each bottleneck dimension, each with a unique seed. This allows the estimation of an ensemble error. The CNN and flow were trained jointly. Both models were trained using an optimizer, a scheduler and a joint scaler. The optimization method chosen was AdamW, the scheduler method was ReduceLROnPlateau and torch's GradScaler ensured more stable backpropagation.

The dataset consisted of 5000 lightcones simulated with 21cmFast, cite the paper somewhere! and the calculated neutral hydrogen fraction as labels. It is split into a training, validation and test set with relative sizes 0.7, 0.2 and 0.1, respectively. Each lightcone is originally $140 \times 140 \times 2350$ voxels in size, corresponding to these redshifts and this space element. In preprocessing, the temperature fluctuation values are normalized individually for each lightcone why is this better? and the lightcone is truncated to $140 \times 140 \times 1916$ voxels, corresponding to a new redshift range of add here. The labels are originially this range in size and are binned to 15-dimensional vectors. They're also normalized. This step reduces the dimensionality of the inference problem, the motivation the mutual information estimation thereby being more tractible.

The hyperparameters used for training are listed in the appendix.

This section might be a good place to write about what is expected in the results

Put GPU used here

9 Results

Does MI increase with accuracy on validation set? Does it increase with training time?

In total, these model where trained. The average runtime for training is insert average runtime.

Figures [2–5] show the training and validation losses for each ensemble the model was trained on. Vertical lines indicate the epoch with best validation performance. Quantify epoch range with best validation performance. Each model then reaches a plateau in epoch range... Does this epoch range change across bottleneck dimensions? I.e. were does model take longer to learn?

Comparability of loss curves? Since loss scaled with bottleneck dim

One can observe a higher seeding variance for models with smaller bottleneck dimension in training and validation losses. Quantify the seed variance for best validation loss here. This indicates a higher training instability for models with smaller bottleneck dimension.

Figures [6–9] show the estimated conditional entropy per epoch for each ensemble the model was trained on. Vertical lines again mark the epoch with best validation performance. The graphs' trends mirror those of the validation losses Do they mirror valdidation losses? This would be expected., and all models reach a saturated estimate of the mutual information in the epoch range... The conditional entropy curves showcase a higher instability than the validation curves Can I quantify this?, which is here

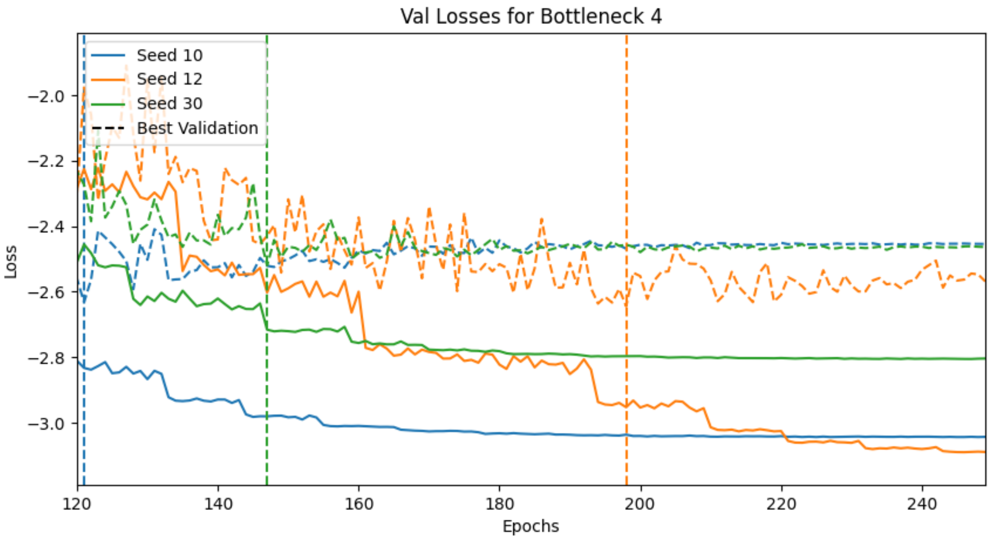


Figure 2: Train and Validation Losses for Bottleneck Dimension 4

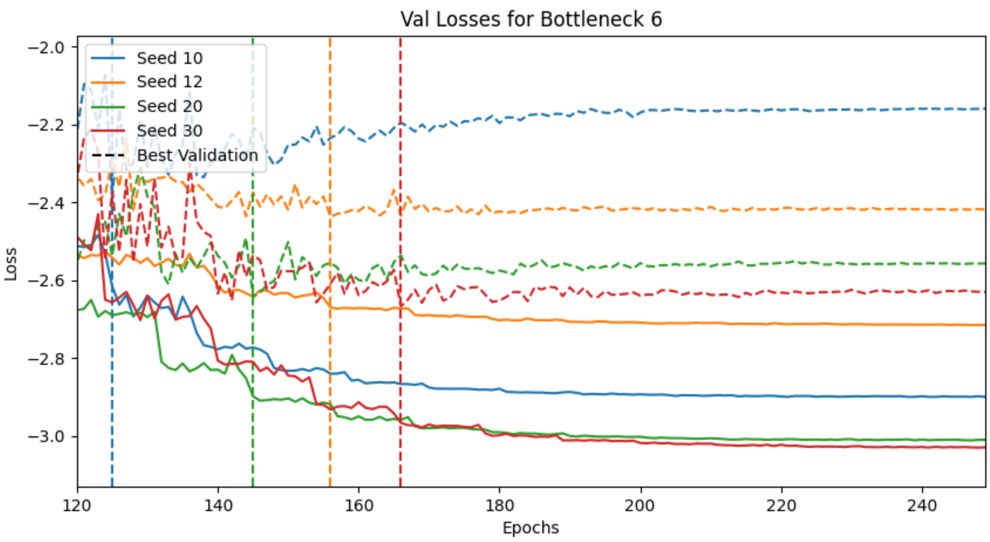


Figure 3: Train and Validation Losses for Bottleneck Dimension 4

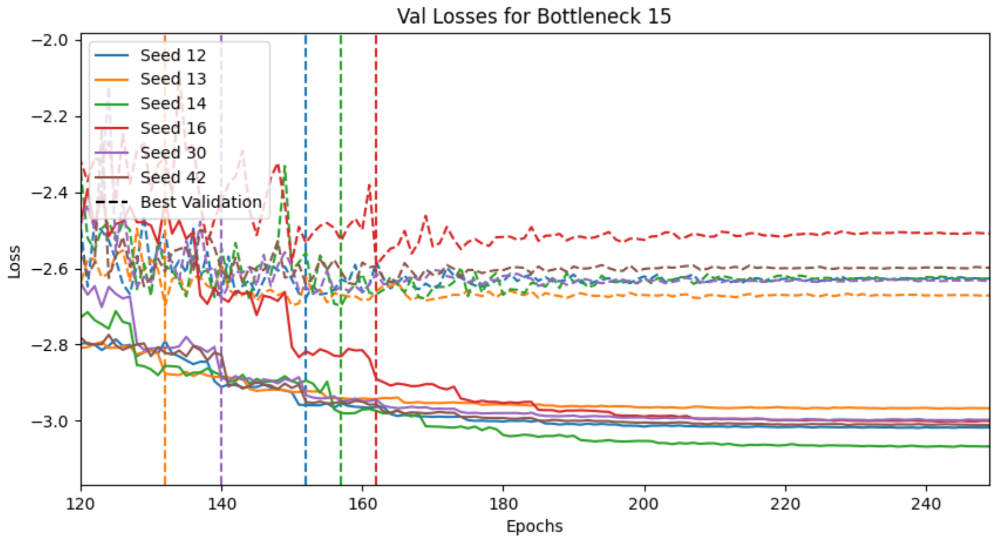


Figure 4: Train and Validation Losses for Bottleneck Dimension 4

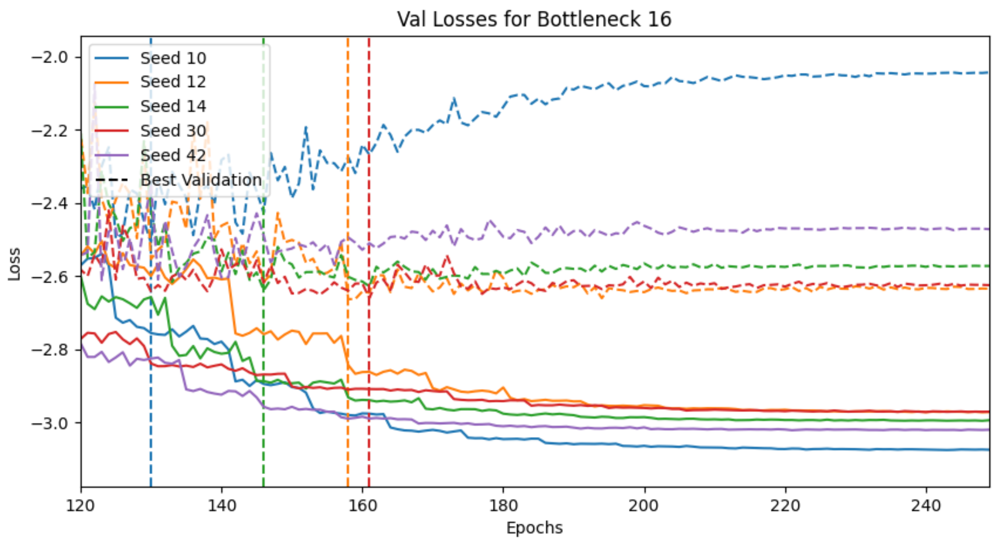


Figure 5: Train and Validation Losses for Bottleneck Dimension 4

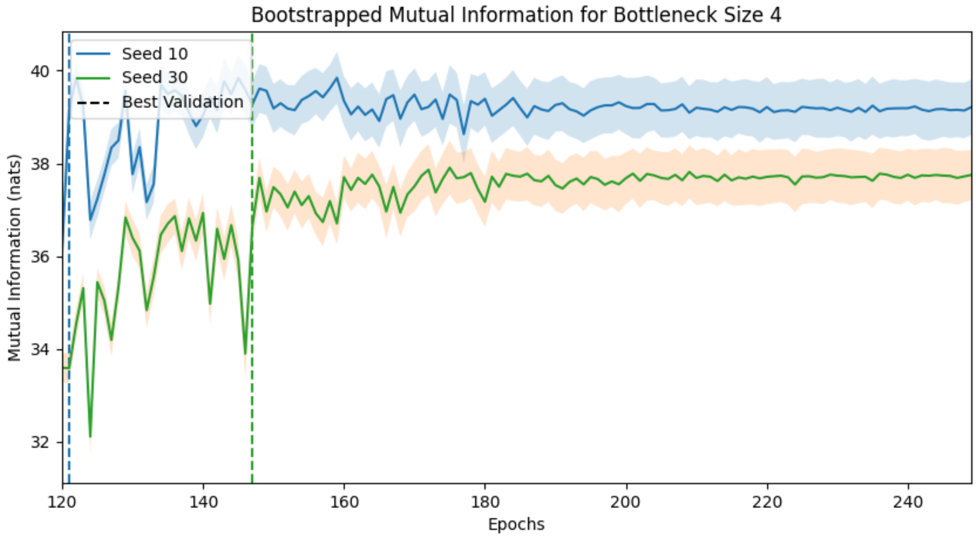


Figure 6: Mutual Information per Epoch for Bottleneck Dimension 4

attributed to the smaller size of the test set (put this interpretation somewhere else?).

The per-epoch estimation of conditional entropy comes with an associated error, here called the *model error*. This is estimated from the distribution of the flow's log-probability of the testing set. Begründen, maybe something from MCMC. According to the CLT, one would expect the error to be estimated by the standard deviation. In this case, the log-probability values are distributed with heavy left-sided kurtosis, which inflates the value of the standard deviation. As an alternative estimate of error, a bootstrapping method is implemented. Quote some source that cites this as best practice. The exact values used for bootstrapping for reproducibility. Something on interpretation of bootstrap results.

For each bottleneck dimension, the final conditional entropy estimate is made by selecting the model with best validation loss from each seed and averaging the conditional entropy for each of these models. The final error on the CE estimate is comprised of the model error and the seed error, stemming from the variability of results across different seeds. The final estimates with errors as shown in table 3. These results can be described as follows: One does observe a general trend of increasing mutual information for increased bottleneck dimension, though the increase is not monotonous. In all cases, the seed error is larger than the model error, indicating training instablity to be the largest source of error in this setup.

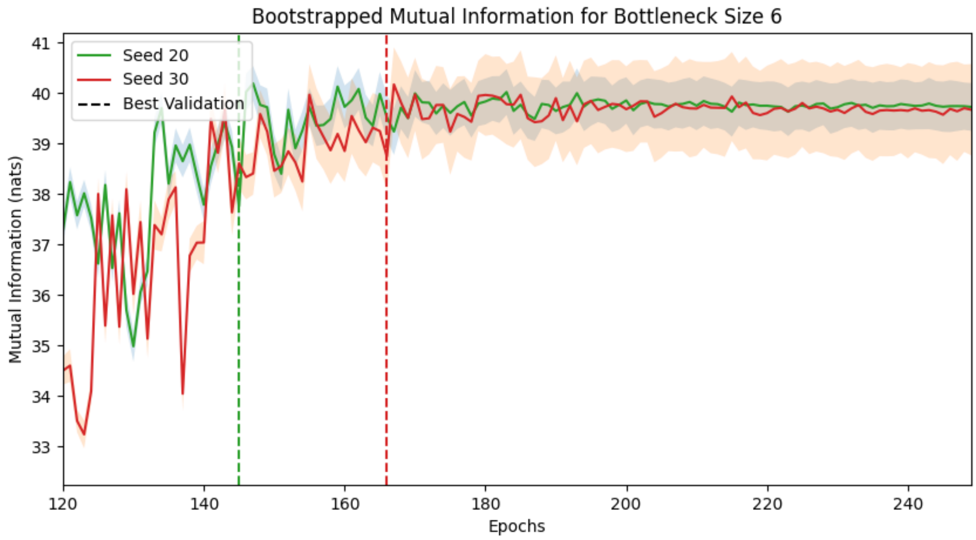


Figure 7: Mutual Information per Epoch for Bottleneck Dimension 4

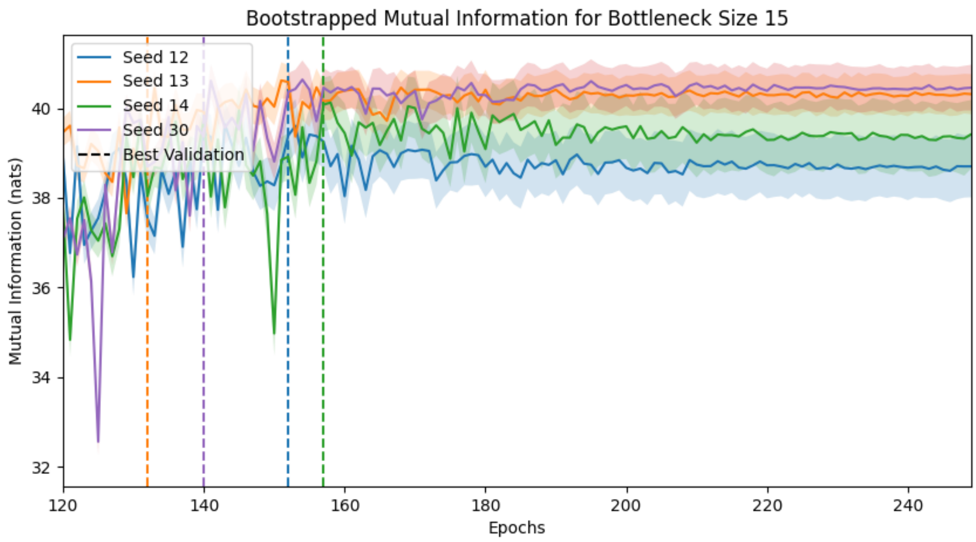


Figure 8: Mutual Information per Epoch for Bottleneck Dimension 4

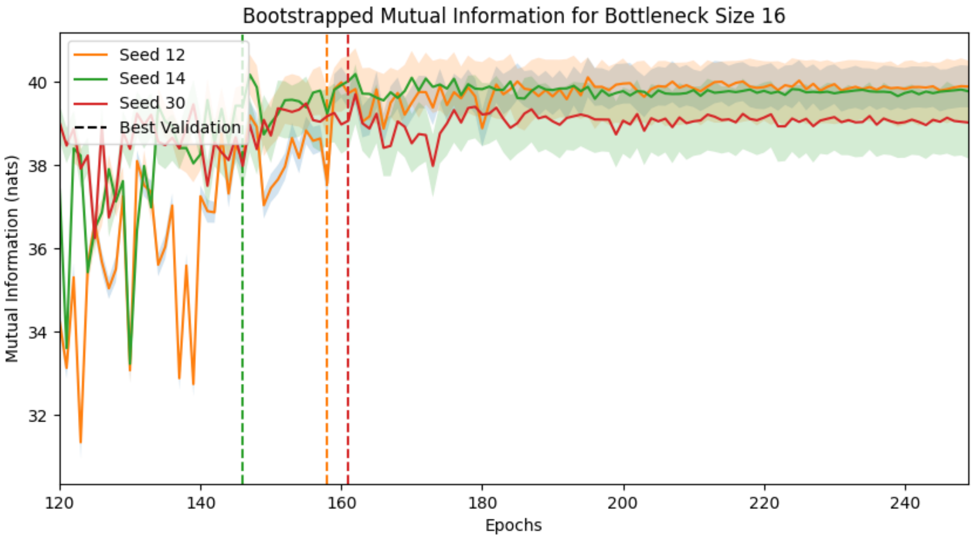


Figure 9: Mutual Information per Epoch for Bottleneck Dimension 4

Table 3: Final Esimation of CE per Bottleneck Dimension (all entropy values in nats)

Bottleneck	CE	Seed error	Model error	No. samples
4	38.81 ± 0.71	0.39	0.12	3
6	38.49 ± 0.95	0.83	0.06	4
15	39.90 ± 0.42	0.13	0.05	5
16	39.57 ± 0.46	0.13	0.09	4

This indicates the seed error being overestimated? Maybe have to exclude some model. Talk about decision mechanisms and how this introduces bias

Results if including selection bias

10 Interpretation

Possible interpretations of the results:

The flow seems to be overcondident, because of the long negative tail in log probability plot. If choosing MI estimation from earlier epochs, get smaller error!

Could overconfidence be due to the choice of priors?

Alternative interpretation:

The CNN learns the latent space. If the weights of the CNN aren't updated enough and the flow is doing too much of the heavy lifting, then the latent space structure might be hard to estimate, resulting in long negative tail.

Check to see wether the extreme models are the reason for this, or if this is a problem with the flow.

The seed uncertainty for this model is very high, given some numbers here. The high sensitivity towards weight initialization points towards unstable training.

Once possible explanation for this instability could be the choice of hyperparameters. For example, a large learning rate might magnify an initially chaotic training process, causing the model to end in different minimia of the loss landscape.

Another possibility is the choice of architecture. For example, the flow might be too deep given the data. While one might expect the need for a deep architecture to infer 15-dimensional labels, these labels are likely highly correlated across dimensions because of curve shape etc, making the problem easier than naively expected. The fact that one observes an early saturation in mutual information across bottleneck sizes supports the latter statement.

Of course, there is always the possibility of there being too little training data. Choosing larger parts of the training set for validation and testing indeed showed an increase in overfitting towards the end of training.

Something about consistency of MI estimate, since e.g. CE estimate is not larger zero from beginning. Maybe cite from paper?

From paper: ”In par- ticular, higher estimated MI between observations and learned representations do not seem to indi- cate improved predictive performance when the representations are used for downstream supervised learning tasks (Tschannen et al., 2019).”

This is also reflected in these data, where MI is sometimes higher! In case not making mistake in indexing in results code.

11 Ausblick

A Network Implementation

Network architecture, training details, losses

References

[1]S. Navas et al. (Particle Data Group), Phys. Rev. D **110**, 030001 (2024).